

Maxwell's Equations, Differential Form

PY 502 — Electromagnetism

The four equations

Gauss's law, no magnetic monopoles, Faraday's law, and the Ampère-Maxwell law, in differential form.

The wave equation falls out

Combining Faraday and Ampère-Maxwell in vacuum gives a wave equation for E with speed $c = 1/\sqrt{\mu_0 \epsilon_0}$.